

Math 142, S14

Quiz 8

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Determine whether the series is convergent or divergent using the Integral Ttest.

$$(a) \sum_{n=1}^{+\infty} \frac{1}{(n+1)^3}$$

$$\begin{aligned} f(x) &= \frac{1}{(x+1)^3} \Rightarrow f(n) = \frac{1}{(n+1)^3}, \\ \int_1^{+\infty} \frac{1}{(x+1)^3} dx &= \lim_{t \rightarrow +\infty} \int_1^t \frac{1}{(x+1)^3} dx \\ &= \lim_{t \rightarrow +\infty} \left[-\frac{1}{2(x+1)^2} \right]_{x=1}^{x=t} \\ &= \lim_{t \rightarrow +\infty} \left[-\frac{1}{2(t+1)^2} + \frac{1}{8} \right] \\ &= \frac{1}{8} \\ \Rightarrow \sum_{n=1}^{+\infty} \frac{1}{(n+1)^3} &\text{ Converges.} \end{aligned}$$

$$(b) \sum_{n=2}^{+\infty} \frac{1}{n \ln n}$$

$$\begin{aligned} f(x) &= \frac{1}{x \ln x} \Rightarrow f(n) = \frac{1}{n \ln n} \\ \int_2^{+\infty} \frac{1}{x \ln x} dx &= \lim_{t \rightarrow +\infty} \int_2^t \frac{1}{x \ln x} dx \\ &= \lim_{t \rightarrow +\infty} \ln(\ln x) \Big|_{x=2}^{x=t} \\ &= \lim_{t \rightarrow +\infty} (\ln(\ln t) - \ln(\ln 2)) \\ &= +\infty \\ \Rightarrow \sum_{n=2}^{+\infty} \frac{1}{n \ln n} &\text{ diverges.} \end{aligned}$$

Problem 2. (5 points) Determine whether the series is convergent or divergent using the Comparison Test or the Limit Comparison Test.

$$(a) \sum_{n=1}^{+\infty} \frac{\cos^2 n}{n^2}$$

$$\text{let } a_n = \frac{\cos^2 n}{n^2}, b_n = \frac{1}{n^2}$$

$$\text{clearly } \cos^2 n \leq 1, \text{ thus}$$

$$0 < a_n \leq b_n$$

$$\text{Since } \sum_{n=1}^{+\infty} \frac{1}{n^2} \text{ Converges,}$$

$$\sum_{n=1}^{+\infty} \frac{\cos^2 n}{n^2} \text{ also converges by}$$

the comparison test.

$$(b) \sum_{n=1}^{+\infty} \frac{n+1}{2n^3+1}$$

$$\text{let } a_n = \frac{n+1}{2n^3+1}, b_n = \frac{n}{2n^3} = \frac{1}{2n^2}$$

$$\begin{aligned} \lim_{n \rightarrow +\infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow +\infty} \frac{n+1}{2n^3+1} \cdot 2n^2 \\ &= \lim_{n \rightarrow +\infty} \frac{2n^3+2}{2n^3+1} = \lim_{n \rightarrow +\infty} \frac{2+\frac{2}{n^3}}{2+\frac{1}{n^3}} \end{aligned}$$

$$= \frac{2}{2} = 1.$$

$$\sum_{n=1}^{+\infty} \frac{1}{2n^2} = \frac{1}{2} \sum_{n=1}^{+\infty} \frac{1}{n^2} \rightarrow \text{Converges}$$

$$\text{Thus } \sum_{n=1}^{+\infty} \frac{n+1}{2n^3+1} \text{ Converges by the}$$

Limit Comparison test.